

Arduino for Musicians

Alex Grant

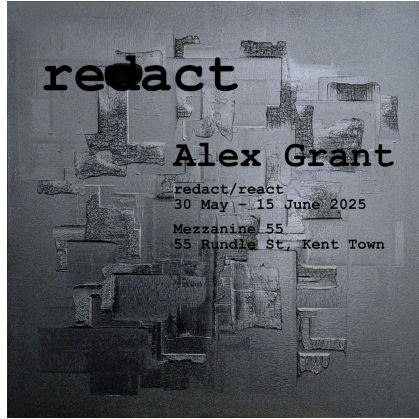
Shameless Plug

redact/react

30 May - 15 June

Mezzanine 55

Solo exhibition



Artist talk 2pm Saturday 7 June

Workshop Plan

Week 1 [Tues, May 13th]

Introduction to Arduino: Focusing on the Arduino microcontroller, participants will be introduced to basics of electronics.



Week 2 [Tues, May 20th]

MIDI: An introduction to the MIDI standard for interconnecting music equipment. Participants will learn how MIDI sends messages for notes and control values. We will build a simple system using the Arduino to send MIDI commands to a digital audio workstation.

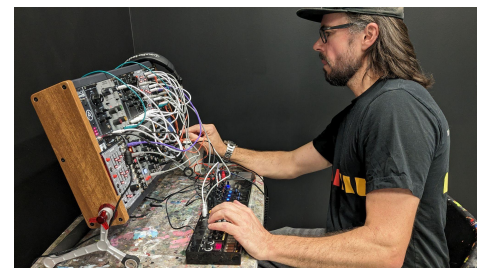


Week 3 [Tues, May 27th]

Modular Synthesis: Using VCV Rack participants will explore sound design and music production using modular synthesis.

Week 4 [Tues, June 3rd]

Homebrew MIDI Controllers: Bringing together the previous three workshops, in this session, participants will build custom MIDI controllers using Arduino and various sensors.



Prerequisites

Week 1 [Tues, May 13th]

- Laptop with [Arduino IDE](#) installed
- Optional: [fritzing](#)
- USB-A to USB-C converter or hub if needed
- Arduino starter kit (provided)



Week 2 [Tues, May 20th]

- Laptop with [Ableton Live](#) and [MIDIView](#) installed
- Alternatively [Garageband](#), [Tracktion](#) or whatever DAW
- Headphones
- Arduino starter kit



Week 3 [Tues, May 27th]

- Laptop with [VCV Rack](#) Installed
- Headphones



Week 4 [Tues, June 3rd]

- Laptop with Arduino IDE, Ableton Live, VCV Rack and MIDIView installed
- Headphones
- Arduino starter kit



Lab Guest
TheLab2024?

Download the Code



<https://github.com/grantaj/labrats-arduino>

git is a tool for tracking changes in files

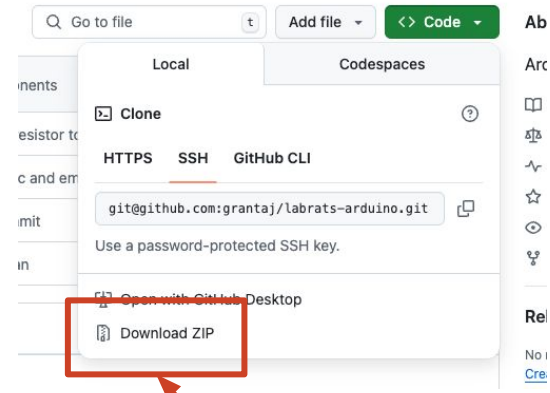
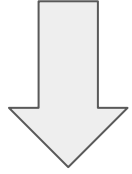
Saves versions of work so you can go back to earlier stages

You can work on different features or ideas in parallel using "branches"

Git works locally, but you can also use websites like **GitHub** to share your work and collaborate

Helps everyone stay organized and avoid overwriting each other's changes

Every change you save with Git is called a "commit", each commit has a message describing the change.



download the code as a .zip archive to your computer

Homebrew MIDI Controllers

Outline

Connecting Arduino to VCV Rack

Switch & Pot

Light Dependent Resistor

3-Axis Accelerometer

Connecting Everything

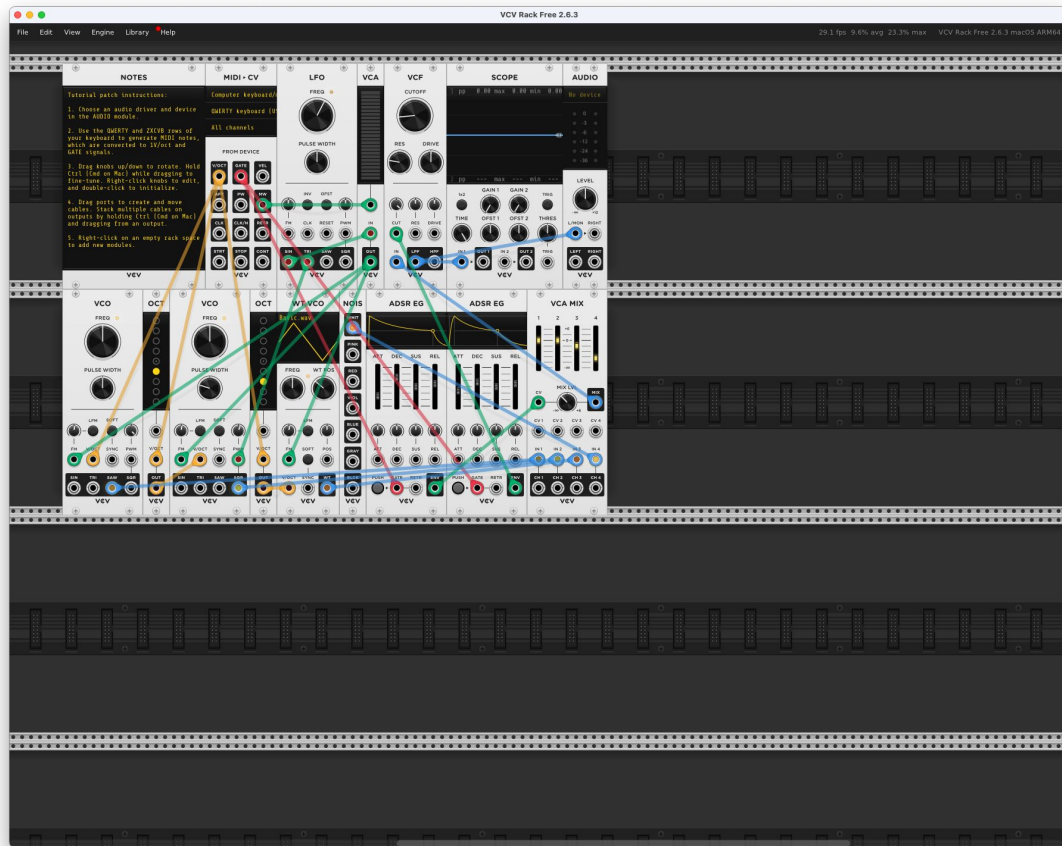
VCV Rack



<https://vcvrack.com/>

Free! (but also paid modules)

Many software versions of
hardware modules



Connect Arduino & Start IDE

Connect Arduino via USB

Start Arduino IDE



Select Arduino Leonardo



Install USB-MIDI Library

Sketch → Include Library → Manage Libraries...

Search for “**USB-MIDI**”

Install

Outline

Connecting Arduino to VCV Rack

Switch & Pot

Light Dependent Resistor

3-Axis Accelerometer

Connecting Everything

Install Bounce2 Library

Sketch → Include Library → Manage Libraries...

Search for “**Bounce2**”

Install

This handles the problem of switch “**bounce**” where pressing the button might trigger several close/open events due to mechanical bouncing.

Debouncing works by ignoring rapid, repeated input signals for a short period after the first signal, ensuring only a single, clean input is registered.

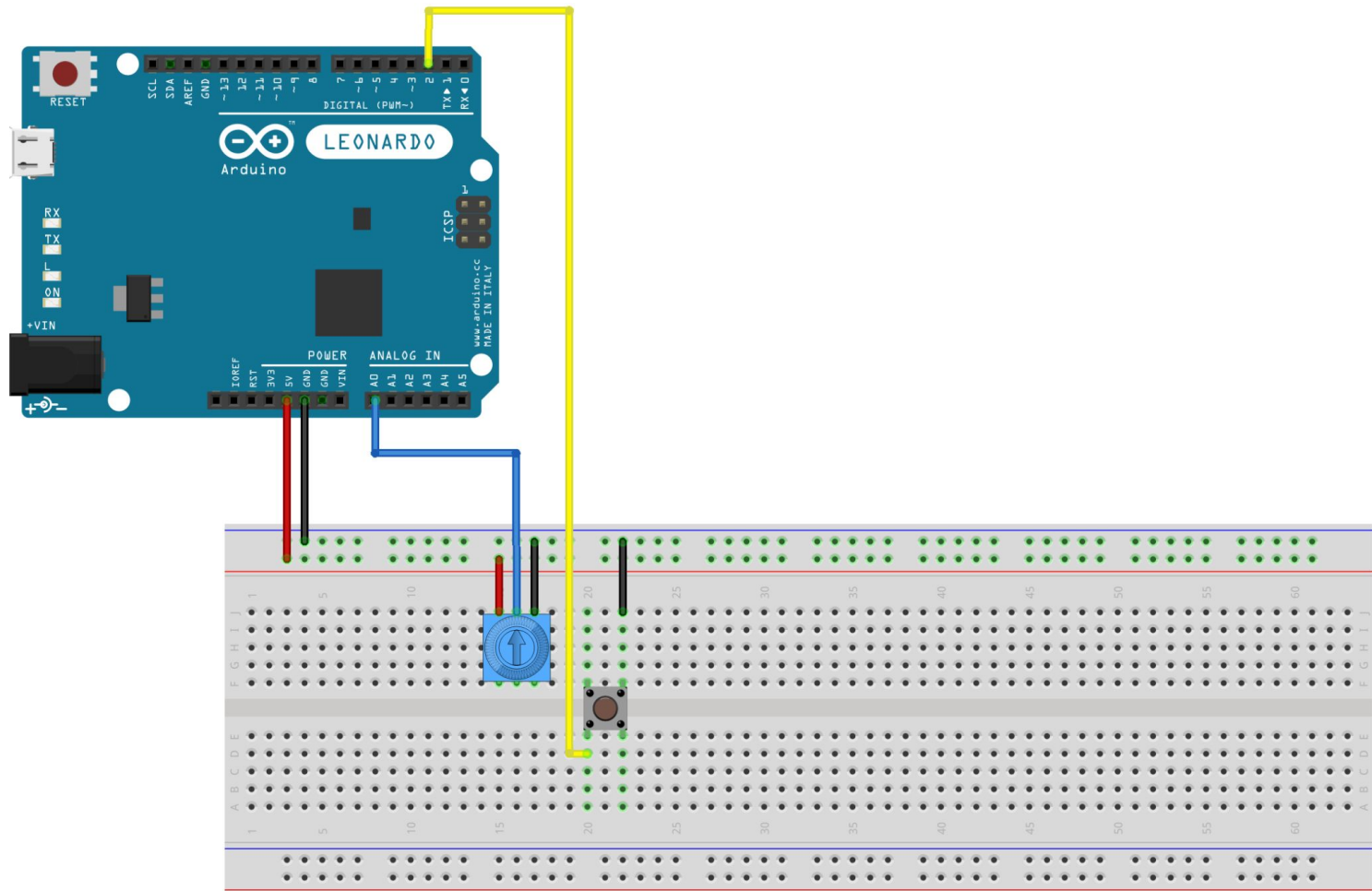
Install ResponsiveAnalogRead Library

Sketch → Include Library → Manage Libraries...

Search for “**ResponsiveAnalogRead**”

Install

Smooths noisy analog readings using adaptive filtering to reduce jitter



Arduino USB MIDI to VCV Rack

Week-4-Homebrew/sketches/2-switch-pot

Week-4-Homebrew/patches/1-basic-voice.vcv

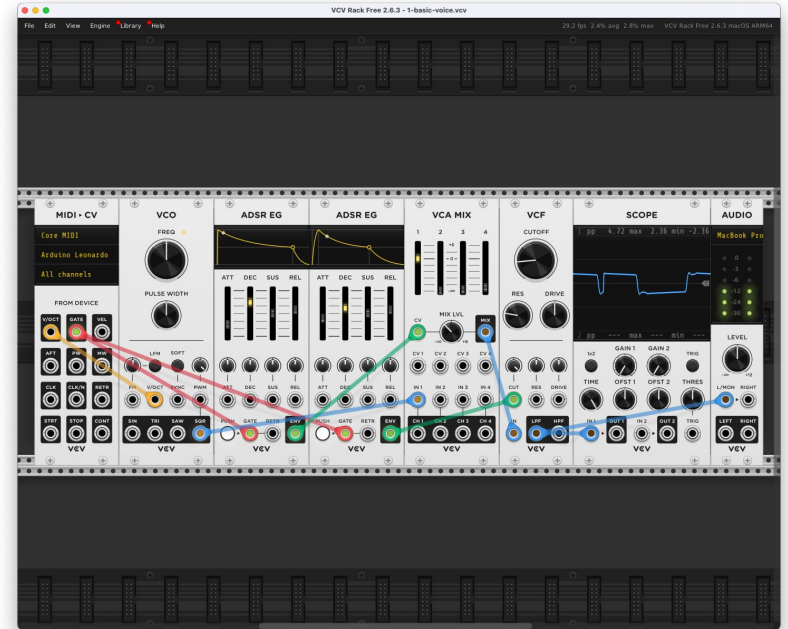
Switch controls note on/off (debounced using Bounce2 library)

Pot controls note value (analog data cleaned up using ResponsiveAnalogRead library)

Note value is going to V/Oct

Note on/off is going to GATE

Use MidiView to spy on MIDI messages



Outline

Connecting Arduino to VCV Rack

Switch & Pot

Light Dependent Resistor

3-Axis Accelerometer

Connecting Everything

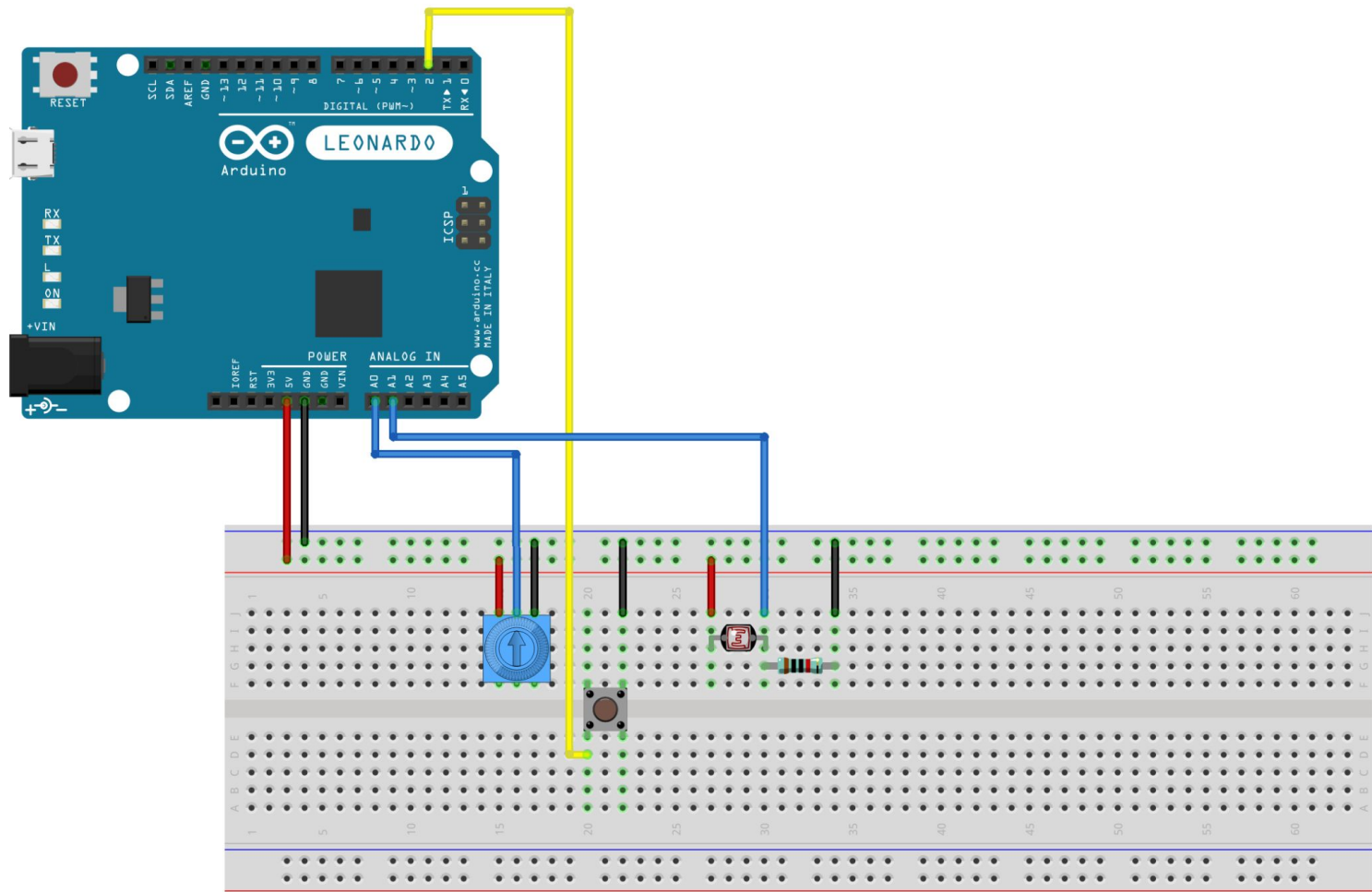
Light Dependent Resistor

Variable resistance, like a potentiometer

Resistance varies according to amount of light rather than by turning a knob

Also need a 10k resistor (brown/black/black/red) to make a voltage divider





Arduino USB MIDI to VCV Rack

Week-4-Homebrew/sketches/3-ldr

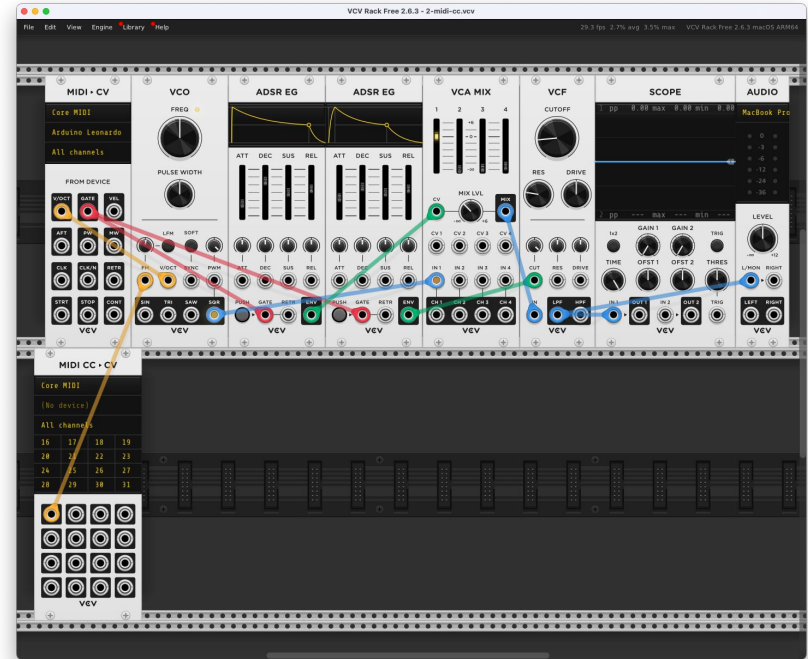
Week-4-Homebrew/patches/3-midi-cc.vcv

MIDI CC - CV module maps incoming MIDI CC values to CVs.

You can click on the numbers and type in new assignments.

LDR is mapped to CC16 in the sketch

Route CV to interesting destinations...



Outline

Connecting Arduino to VCV Rack

Switch & Pot

Light Dependent Resistor

3-Axis Accelerometer

Connecting Everything

Install MPU6050 Library

Sketch → Include Library → Manage Libraries...

Search for “**Adafruit MPU6050**”

Install

Provides easy methods to read the accelerometer

3 Axis Accelerometer

MPU6050 3-axis accelerometer and gyroscope

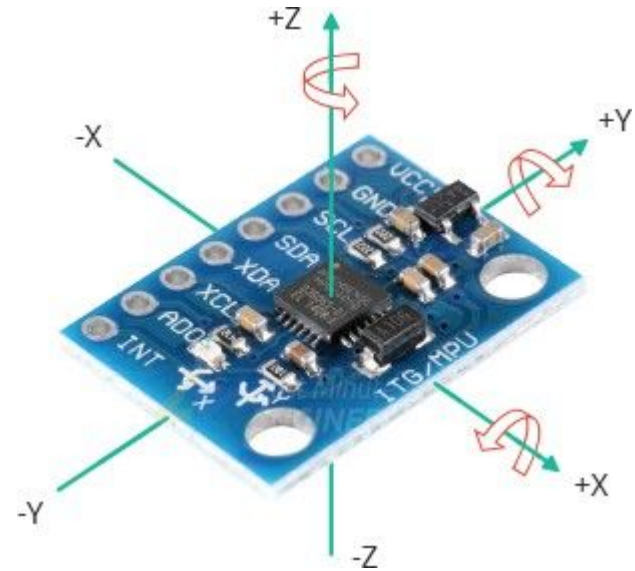
Measures acceleration in x/y/z directions

Since gravity is acceleration (9.8m/s/s) this is equivalent to measuring orientation

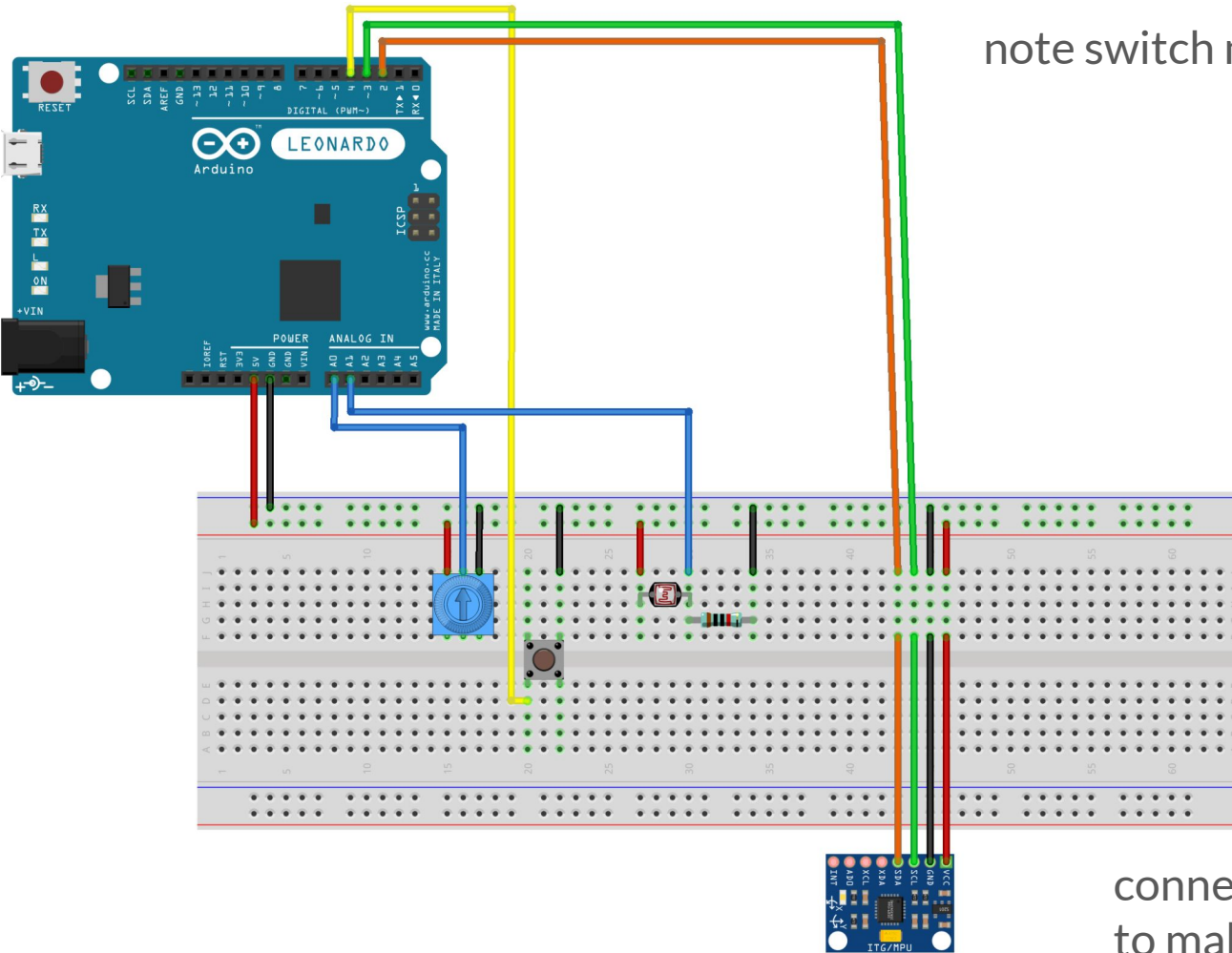
Also measures rotational speed around each axis

Communicates using I2C protocol (two wires)

I2C is a common protocol to send digital data to and from microcontrollers



note switch moved to pin 4



Arduino USB MIDI to VCV Rack

Week-4-Homebrew/sketches/4-accelerometer

Week-4-Homebrew/patches/4-midi-cc-2.vcv

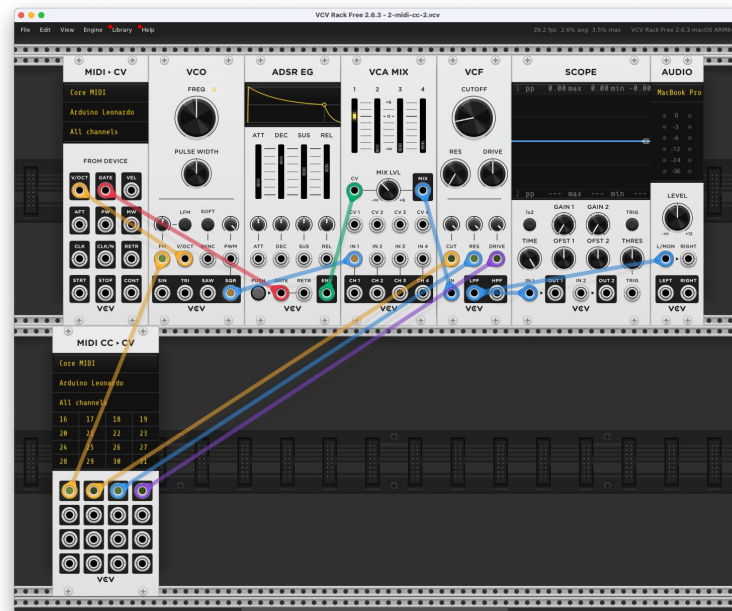
MIDI CC - CV module maps incoming MIDI CC values to CVs.

You can click on the numbers and type in new assignments.

LDR is mapped to CC16 (patched to frequency modulation)

Accelerometer is mapped to 17/18/19 (patched to cutoff/resonance/drive on the VCF)

Route CV to interesting destinations...



Outline

Connecting Arduino to VCV Rack

Switch & Pot

Light Dependent Resistor

3-Axis Accelerometer

Ultrasonic Distance Sensor

Install NewPing Library

Sketch → Include Library → Manage Libraries...

Search for “**NewPing**”

Install

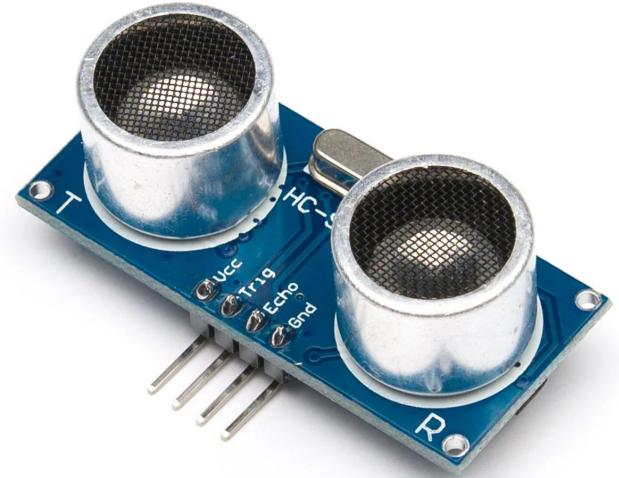
Provides easy methods to read the ultrasonic sensor

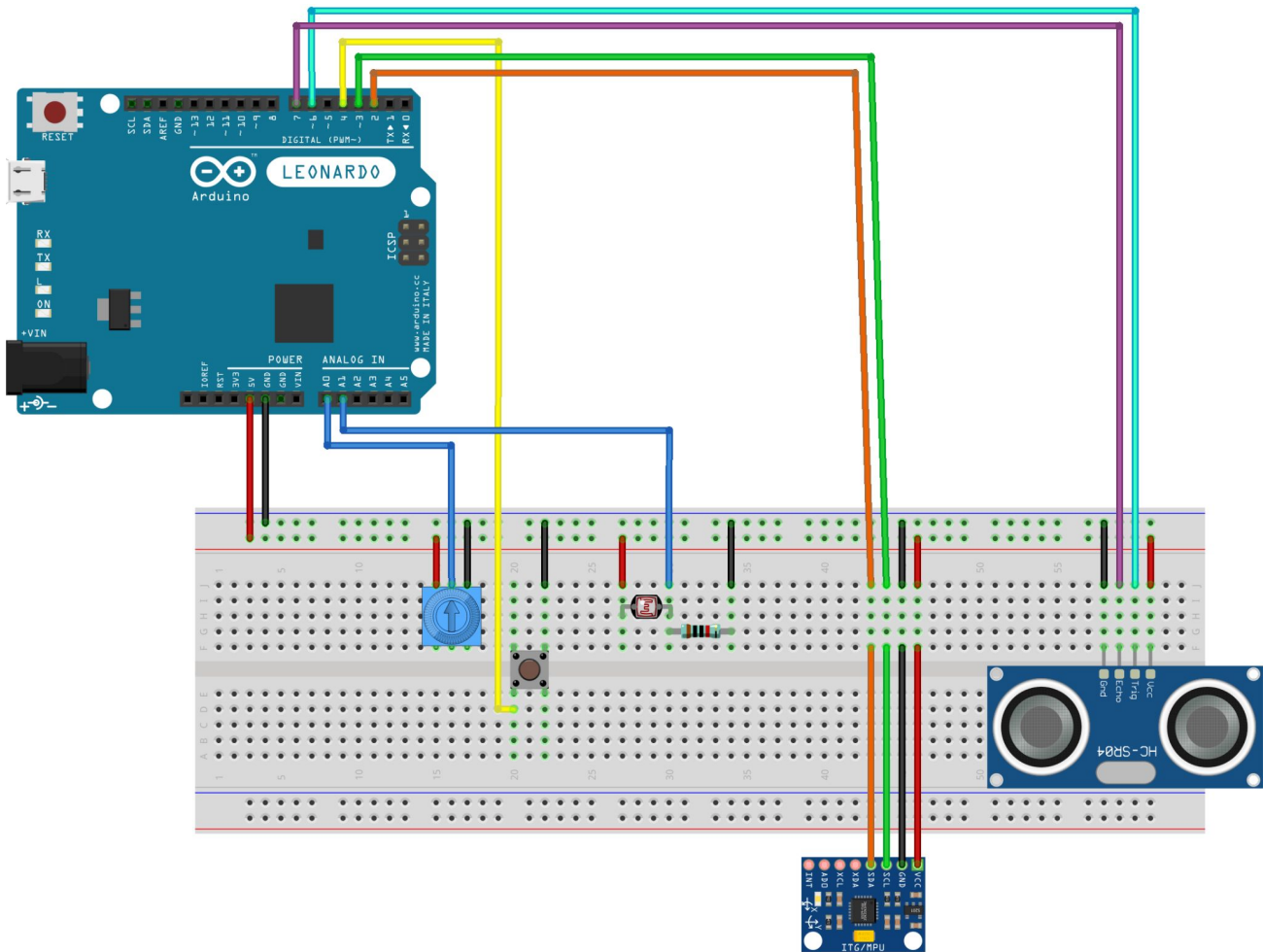
Ultrasonic Sensor

HC SR-04

Sends an ultrasonic pulse from its trigger pin and measures the time it takes for the echo pin to receive the reflected signal.

By calculating the time delay, can determine distance, typically ranging from 2 cm to 400 cm.





Arduino USB MIDI to VCV Rack

Week-4-Homebrew/sketches/5-ultrasonic

Week-4-Homebrew/patches/5-midi-cc-3.vcv

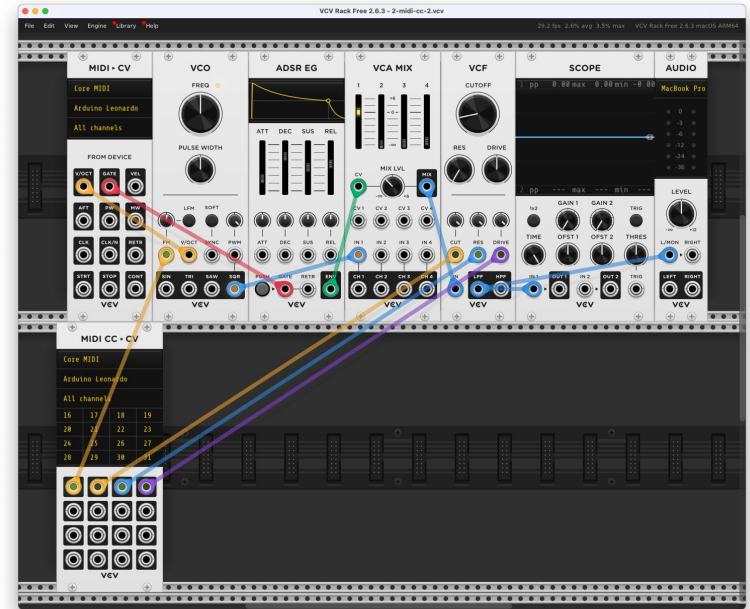
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Thank You!